

CHS Linear and Logistic Charts

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```
chs_data <- read.csv("CHS_cohortE(in).csv")
str(chs_data)

## 'data.frame':    1000 obs. of  25 variables:
## $ id             : int  54577 50863 52081 53817 54683 55339 55766 51056 5491
## 9 52992 ...
## $ townabbr       : chr   "SA" "SD" "SD" "RV" ...
## $ age            : num   15.1 16.5 15.6 15.2 14.2 15.2 15.8 16 15.2 16.1 ...
## $ male           : int    1 1 0 1 1 0 1 1 0 0 ...
## $ race           : chr   "Others" "Mixed" "Caucasian" "Unknown or Missing" ...
## $ hisp           : chr   "Hispanic" "Hispanic" "Non-Hispanic" "Hispanic" ...
## $ asthma         : int    0 NA 0 0 1 0 0 1 0 0 ...
## $ height         : int   168 168 167 160 169 161 185 183 163 165 ...
## $ weight         : num   52 50.2 55.6 60.9 62.1 ...
## $ bmi            : num   18.4 17.8 19.9 23.8 21.8 20.6 33.2 39 28.6 28.4 ...
## $ educ           : int    1 1 3 2 5 5 2 3 1 2 ...
## $ HomeBuilt      : chr   "1980 or later" "Unknown or Missing" "1960s to 1970s
## " "Unknown or Missing" ...
## $ BaseGasstove   : int    1 0 1 1 1 0 1 1 1 1 ...
## $ BasePets       : int    1 0 1 0 1 1 1 1 1 0 ...
## $ ETS_base       : int    0 0 0 0 0 0 0 1 0 0 ...
## $ wheeze         : int    0 NA 0 0 0 0 0 0 0 0 ...
## $ fev1           : int  4090 3790 3240 3890 3730 3530 5420 4480 3290 3390 ...
## $ fvc            : int  4950 4810 3370 4190 4930 4010 6360 5590 3450 3930 ...
## $ pm25           : num    8.84 14.28 15 15.76 14.18 ...
## $ sulfate        : num    0.93 1.38 1.46 1.57 1.32 ...
## $ nitrate        : num    1.87 2.28 2.48 2.45 2.18 ...
## $ ec             : num    0.702 0.873 0.884 0.762 0.893 ...
## $ dust           : num    0.449 1.302 1.246 1.29 1.34 ...
## $ longitude      : num   -120 -118 -118 -117 -118 ...
## $ latitude       : num    34.5 34.1 34.1 34 34.1 ...

## Convert Binary/Categorical Variables first

library(dplyr)

## Warning: package 'dplyr' was built under R version 4.6.1

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag
```

```

## The following objects are masked from 'package:base':
##
## intersect, setdiff, setequal, union

chs_data %>%
  dplyr::mutate(across(c(townabbr, male, race, hisp, asthma, educ, BaseGasstove, BasePets, ETS_base, wheeze), ~as.factor(.x)))

str(chs_data)

## 'data.frame': 1000 obs. of 25 variables:
## $ id : int 54577 50863 52081 53817 54683 55339 55766 51056 54919 52992 ...
## $ townabbr : chr "SA" "SD" "SD" "RV" ...
## $ age : num 15.1 16.5 15.6 15.2 14.2 15.2 15.8 16 15.2 16.1 ...
## $ male : int 1 1 0 1 1 0 1 1 0 0 ...
## $ race : chr "Others" "Mixed" "Caucasian" "Unknown or Missing" ...
## $ hisp : chr "Hispanic" "Hispanic" "Non-Hispanic" "Hispanic" ...
## $ asthma : int 0 NA 0 0 1 0 0 1 0 0 ...
## $ height : int 168 168 167 160 169 161 185 183 163 165 ...
## $ weight : num 52 50.2 55.6 60.9 62.1 ...
## $ bmi : num 18.4 17.8 19.9 23.8 21.8 20.6 33.2 39 28.6 28.4 ...
## $ educ : int 1 1 3 2 5 5 2 3 1 2 ...
## $ HomeBuilt : chr "1980 or later" "Unknown or Missing" "1960s to 1970s" "Unknown or Missing" ...
## $ BaseGasstove: int 1 0 1 1 1 0 1 1 1 1 ...
## $ BasePets : int 1 0 1 0 1 1 1 1 1 0 ...
## $ ETS_base : int 0 0 0 0 0 0 0 1 0 0 ...
## $ wheeze : int 0 NA 0 0 0 0 0 0 0 0 ...
## $ fev1 : int 4090 3790 3240 3890 3730 3530 5420 4480 3290 3390 ...
## $ fvc : int 4950 4810 3370 4190 4930 4010 6360 5590 3450 3930 ...
## $ pm25 : num 8.84 14.28 15 15.76 14.18 ...
## $ sulfate : num 0.93 1.38 1.46 1.57 1.32 ...
## $ nitrate : num 1.87 2.28 2.48 2.45 2.18 ...
## $ ec : num 0.702 0.873 0.884 0.762 0.893 ...
## $ dust : num 0.449 1.302 1.246 1.29 1.34 ...
## $ longitude : num -120 -118 -118 -117 -118 ...
## $ latitude : num 34.5 34.1 34.1 34 34.1 ...

## Linear Regression (Continuous Outcomes)

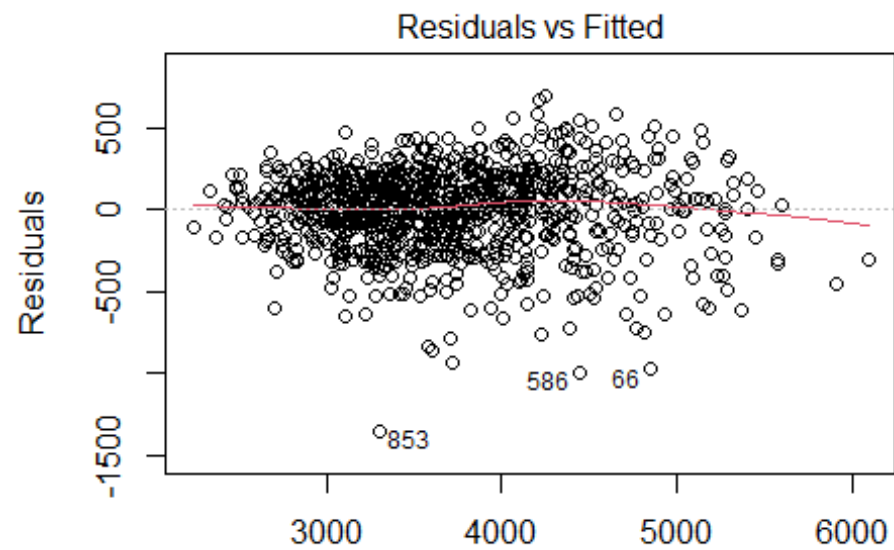
linear_model <- lm(fev1 ~ age + bmi + weight +
  height + HomeBuilt + fvc +
  pm25 + ec + sulfate +
  nitrate + latitude,
  data = chs_data)
summary(linear_model)

##
## Call:
## lm(formula = fev1 ~ age + bmi + weight + height + HomeBuilt +

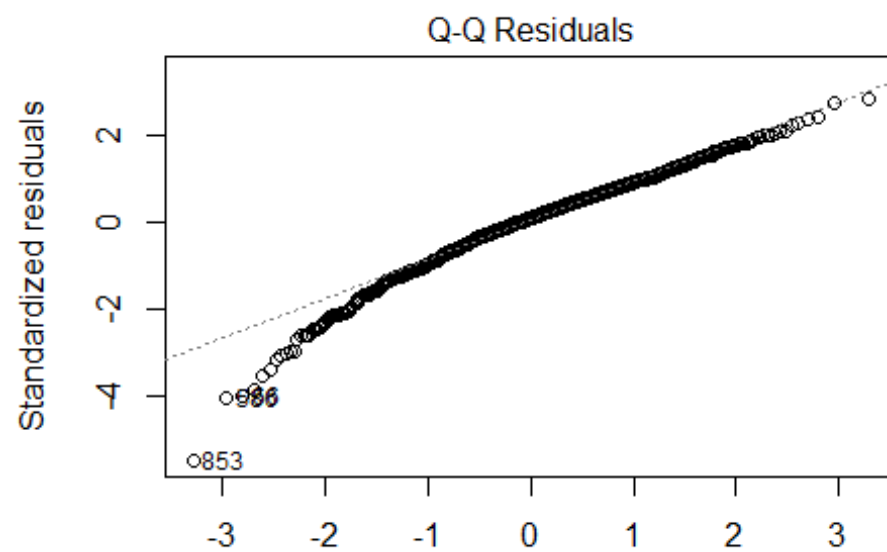
```

```
##      fvc + pm25 + ec + sulfate + nitrate + latitude, data = chs_data)
##
## Residuals:
##      Min        1Q      Median        3Q        Max
## -1350.94  -138.33    24.38    161.78    696.32
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -8.763e+03  3.439e+03  -2.548   0.0110 *
## age            2.963e+01  1.352e+01   2.192   0.0286 *
## bmi            9.218e+00  1.506e+01   0.612   0.5406
## weight        -4.278e+00  5.340e+00  -0.801   0.4232
## height         5.985e+00  4.388e+00   1.364   0.1729
## HomeBuilt1980 or later -2.324e+01  2.299e+01  -1.011   0.3124
## HomeBuiltBefore 1960  -2.900e+01  2.163e+01  -1.341   0.1804
## HomeBuiltUnknown or Missing 2.002e+00  2.488e+01   0.080   0.9359
## fvc            7.397e-01  1.526e-02  48.478 < 2e-16 ***
## pm25           1.764e+01  7.939e+00   2.222   0.0265 *
## ec            -1.917e+02  1.571e+02  -1.220   0.2228
## sulfate       -2.845e+02  1.157e+02  -2.458   0.0142 *
## nitrate        2.034e+02  5.033e+01   4.041 5.74e-05 ***
## latitude       2.270e+02  9.457e+01   2.400   0.0166 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 247.4 on 965 degrees of freedom
## (21 observations deleted due to missingness)
## Multiple R-squared:  0.8749, Adjusted R-squared:  0.8733
## F-statistic: 519.4 on 13 and 965 DF,  p-value: < 2.2e-16

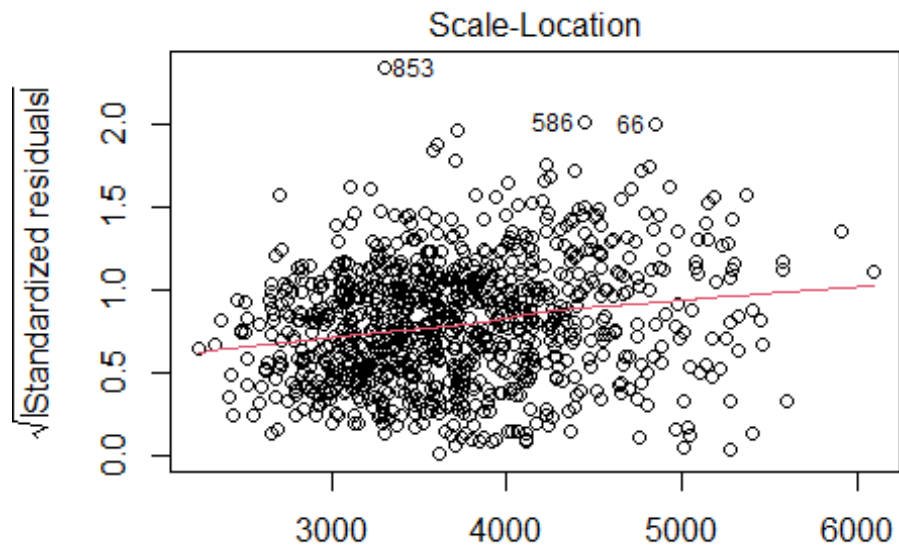
plot(linear_model)
```



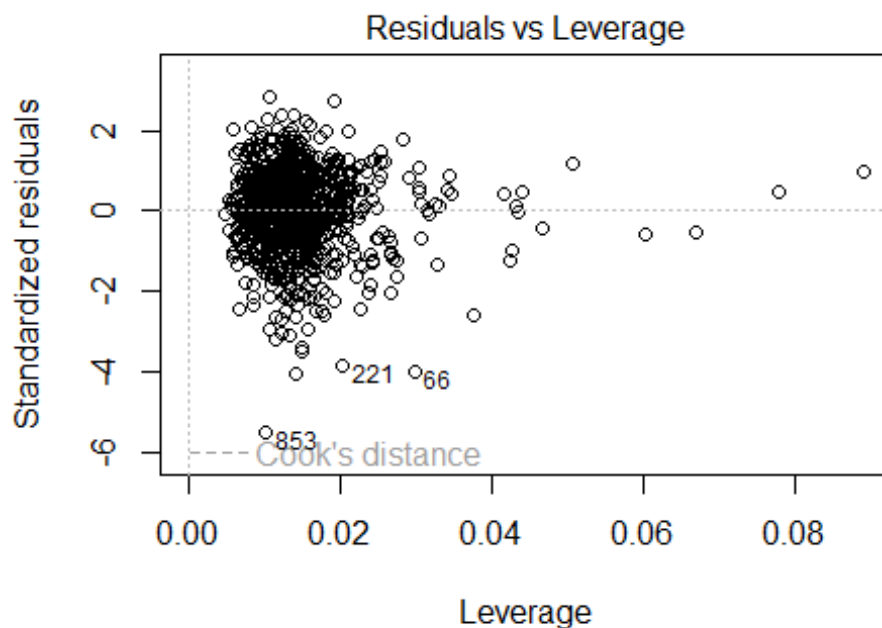
Fitted values
 $\eta(\text{fev1} \sim \text{age} + \text{bmi} + \text{weight} + \text{height} + \text{HomeBuilt} + \text{fvc} + \text{pm25} + \text{ec} +$



Theoretical Quantiles
 $\eta(\text{fev1} \sim \text{age} + \text{bmi} + \text{weight} + \text{height} + \text{HomeBuilt} + \text{fvc} + \text{pm25} + \text{ec} +$



Fitted values
`lm(fev1 ~ age + bmi + weight + height + HomeBuilt + fvc + pm25 + ec +`



`lm(fev1 ~ age + bmi + weight + height + HomeBuilt + fvc + pm25 + ec +`

Logistic Regression (Binary Outcome)

```
logistic_model <- glm(asthma ~ latitude +
  BaseGasstove + BasePets
```

```

+ wheeze + ETS_base + educ +
  hisp + race + male +
  townabbr,
data = chs_data,
family = binomial,
na.action = na.exclude)

```

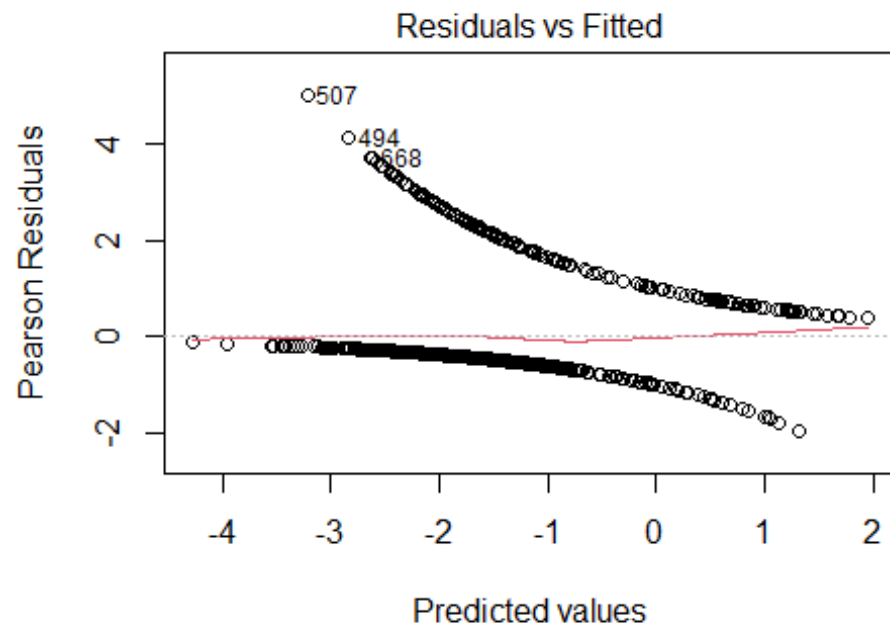
```
summary(logistic_model)
```

```

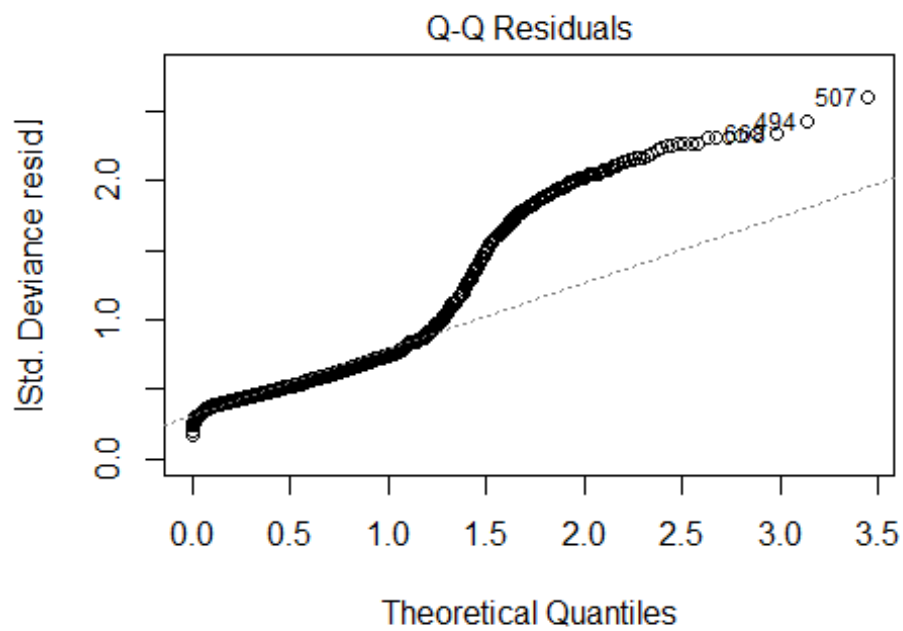
##
## Call:
## glm(formula = asthma ~ latitude + BaseGasstove + BasePets + wheeze +
##      ETS_base + educ + hisp + race + male + townabbr, family = binomial,
##      data = chs_data, na.action = na.exclude)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    87.73466   87.98974   0.997   0.3187
## latitude       -2.64998    2.59947  -1.019   0.3080
## BaseGasstove    0.11564    0.27706   0.417   0.6764
## BasePets       -0.02063    0.19992  -0.103   0.9178
## wheeze         2.32974    0.23969   9.720 <2e-16 ***
## ETS_base       -0.55586    0.30888  -1.800   0.0719 .
## educ           0.17105    0.09465   1.807   0.0708 .
## hispNon-Hispanic -0.06827    0.26990  -0.253   0.8003
## hispUnknown or Missing -1.32512    0.81895  -1.618   0.1056
## raceAsian       0.57173    0.78449   0.729   0.4661
## raceCaucasian   0.03793    0.70400   0.054   0.9570
## raceMixed      -0.21104    0.73972  -0.285   0.7754
## raceOthers     -0.05054    0.75049  -0.067   0.9463
## raceUnknown or Missing -0.26172    0.78616  -0.333   0.7392
## male           0.31103    0.18982   1.639   0.1013
## townabbrGL      0.27611    0.85214   0.324   0.7459
## townabbrLB     -0.74208    0.57588  -1.289   0.1975
## townabbrML     -0.36513    0.60319  -0.605   0.5450
## townabbrRV      0.42174    0.47251   0.893   0.3721
## townabbrSA      0.81115    1.82020   0.446   0.6559
## townabbrSD      0.69673    0.77287   0.901   0.3673
## townabbrUP     -0.24505    0.84400  -0.290   0.7716
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 890.31  on 876  degrees of freedom
## Residual deviance: 742.90  on 855  degrees of freedom
## (123 observations deleted due to missingness)
## AIC: 786.9
##
## Number of Fisher Scoring iterations: 5

```

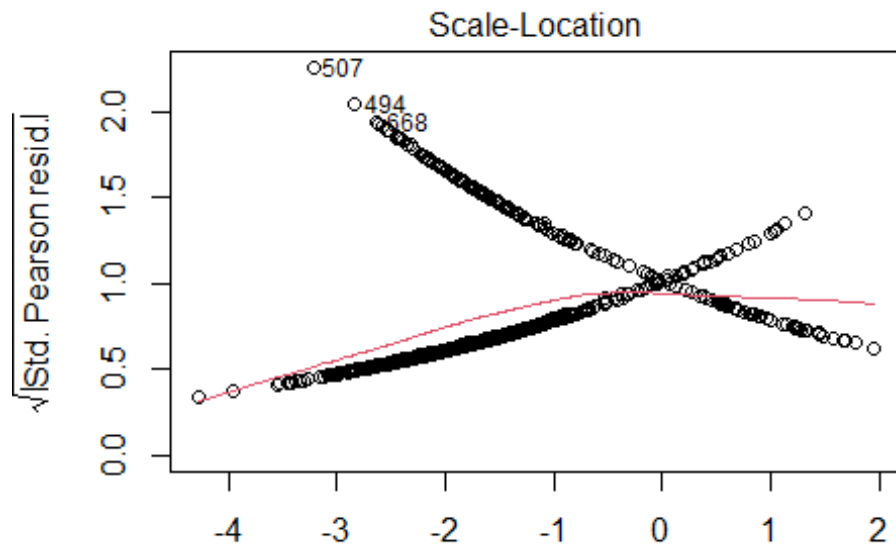
```
plot(logistic_model)
```



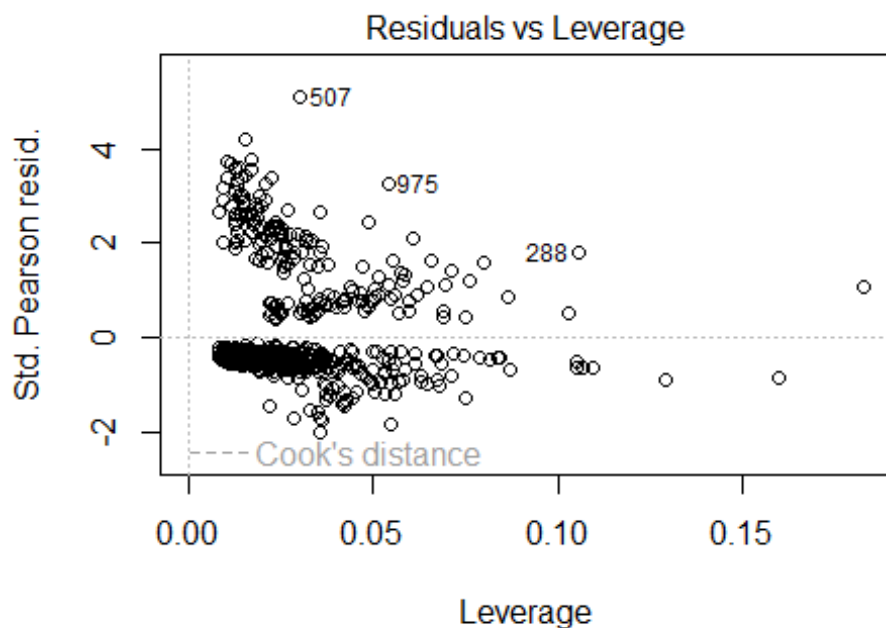
```
isthma ~ latitude + BaseGasstove + BasePets + wheeze + ETS_base
```



```
isthma ~ latitude + BaseGasstove + BasePets + wheeze + ETS_base
```



asthma ~ latitude + BaseGasstove + BasePets + wheeze + ETS_base



asthma ~ latitude + BaseGasstove + BasePets + wheeze + ETS_base

Predicted Probabilities & Odds Ratios (IMPORTANT)

```
chs_data$predicted_prob <- predict(logistic_model, type = "response")
```

```
exp(coef(logistic_model))
```

```
##           (Intercept)           latitude      BaseGasstove
##      1.266709e+38      7.065238e-02      1.122591e+00
##           BasePets           wheeze           ETS_base
##      9.795860e-01      1.027530e+01      5.735809e-01
##           educ      hispNon-Hispanic      hispUnknown or Missing
##      1.186544e+00      9.340124e-01      2.657720e-01
##           raceAsian      raceCaucasian      raceMixed
##      1.771335e+00      1.038657e+00      8.097404e-01
##           raceOthers      raceUnknown or Missing      male
##      9.507134e-01      7.697256e-01      1.364826e+00
##           townabbrGL      townabbrLB      townabbrML
##      1.317995e+00      4.761228e-01      6.941033e-01
##           townabbrRV      townabbrSA      townabbrSD
##      1.524605e+00      2.250487e+00      2.007182e+00
##           townabbrUP
##      7.826677e-01
```

```
chs_data <- na.omit(chs_data)
```

```
## Clean model with ONLY continuous predictors in one model
```

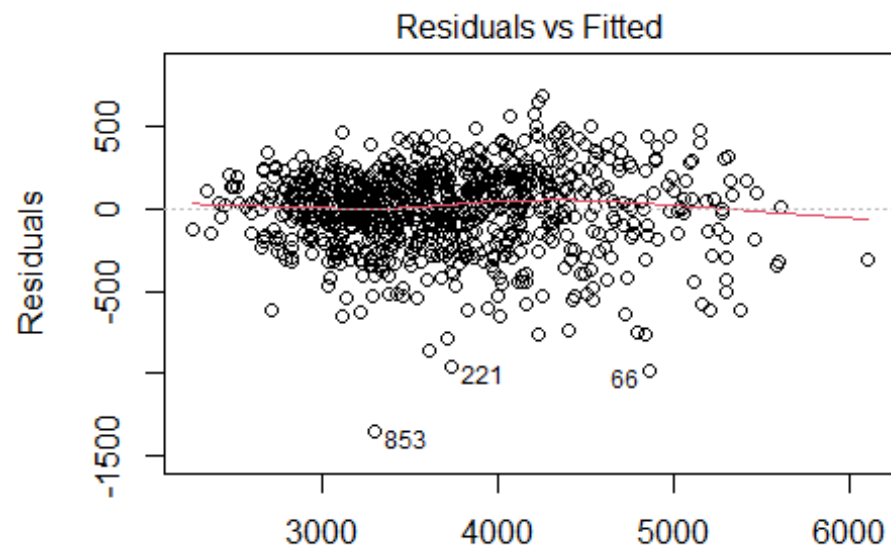
```
linear_model_clean <- lm(fev1 ~ age+ bmi + weight + height +
      HomeBuilt + fvc + pm25 + ec + sulfate
      + nitrate + latitude,
      data = chs_data)
```

```
summary(linear_model_clean)
```

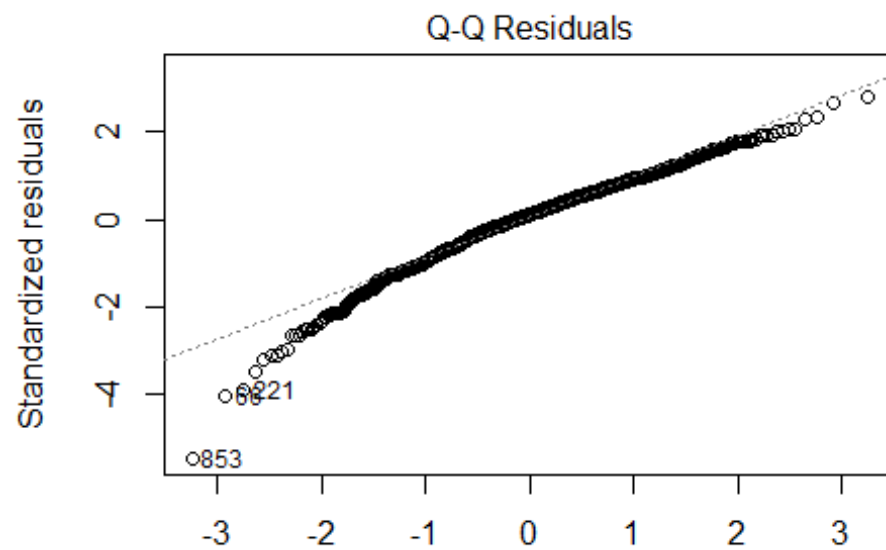
```
##
## Call:
## lm(formula = fev1 ~ age + bmi + weight + height + HomeBuilt +
##      fvc + pm25 + ec + sulfate + nitrate + latitude, data = chs_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1350.07  -145.04    25.95   162.31   686.36
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -7714.5452   3680.4822  -2.096   0.03637 *
## age           29.9867    14.2064    2.111   0.03508 *
## bmi           8.0326     16.2593    0.494   0.62141
## weight       -3.8063     5.7632   -0.660   0.50915
## height        5.6106     4.7039    1.193   0.23330
## HomeBuilt1980 or later -27.1693    23.7558  -1.144   0.25307
## HomeBuiltBefore 1960  -23.8229    22.3039  -1.068   0.28578
## HomeBuiltUnknown or Missing  5.5486    28.2465   0.196   0.84432
```

```
## fvc                0.7429      0.0164  45.312 < 2e-16 ***
## pm25              17.5768      8.4838   2.072  0.03859 *
## ec               -205.9377    163.4358  -1.260  0.20800
## sulfate          -251.4291    125.2184  -2.008  0.04497 *
## nitrate           174.4154     54.6504   3.191  0.00147 **
## latitude          198.6099    101.3652   1.959  0.05040 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 246.7 on 844 degrees of freedom
## Multiple R-squared:  0.8761, Adjusted R-squared:  0.8742
## F-statistic:  459 on 13 and 844 DF,  p-value: < 2.2e-16

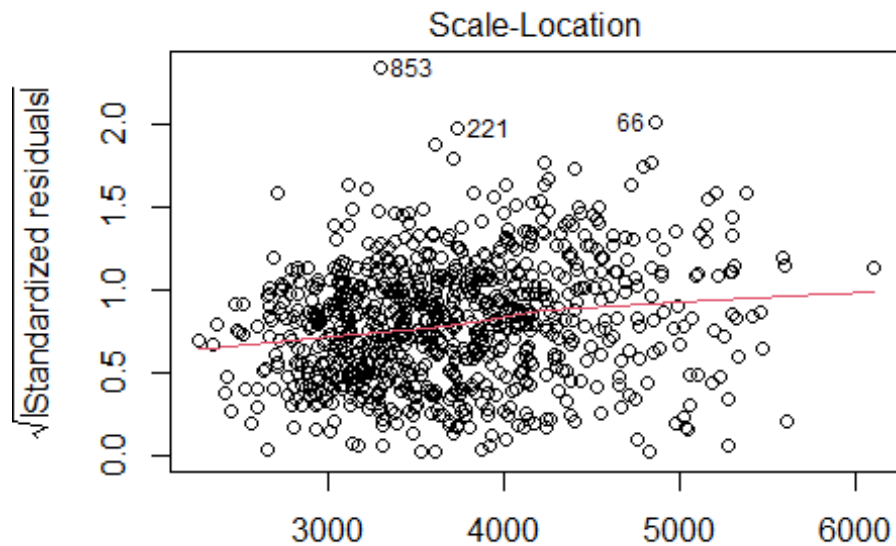
plot(linear_model_clean)
```



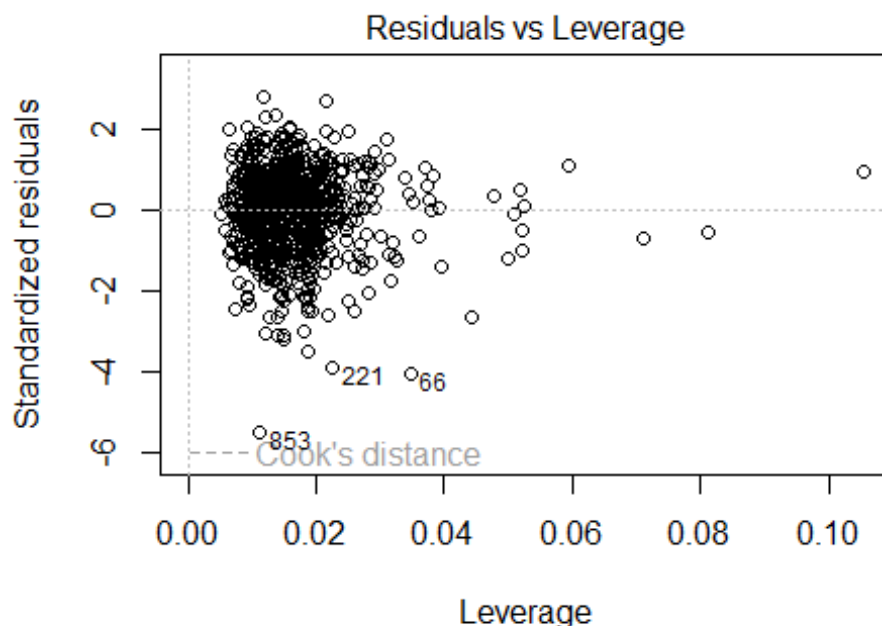
Fitted values
 $\eta(\text{fev1} \sim \text{age} + \text{bmi} + \text{weight} + \text{height} + \text{HomeBuilt} + \text{fvc} + \text{pm25} + \text{ec} +$



Theoretical Quantiles
 $\eta(\text{fev1} \sim \text{age} + \text{bmi} + \text{weight} + \text{height} + \text{HomeBuilt} + \text{fvc} + \text{pm25} + \text{ec} +$



$\eta(\text{fev1} \sim \text{age} + \text{bmi} + \text{weight} + \text{height} + \text{HomeBuilt} + \text{fvc} + \text{pm25} + \text{ec} +$



$\eta(\text{fev1} \sim \text{age} + \text{bmi} + \text{weight} + \text{height} + \text{HomeBuilt} + \text{fvc} + \text{pm25} + \text{ec} +$

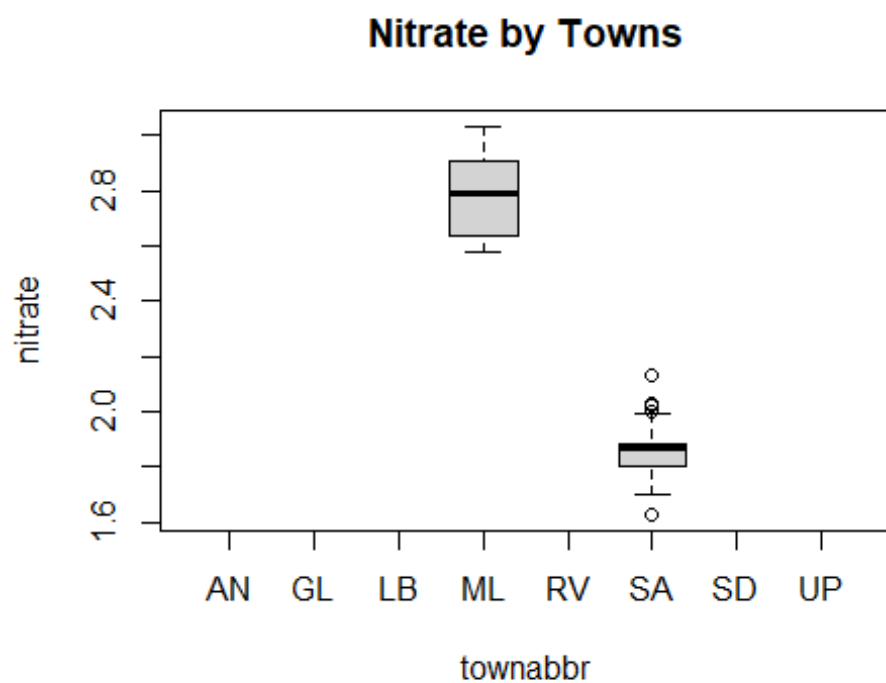
Moving from here we focus on the pollutant variables pm2.5 & nitrate, as we ll as crucial variables that we think plays a role in the found results (age, bmi, educ, male, latitude). Towns: Santa Barbara - having the most amount of subjects tested, Lowest pm2.5 mean(8.68(0.399)), & Lowest mean Nitrate and M

```
## we start by sectioning off the towns we want and running several correlat
ion tests
```

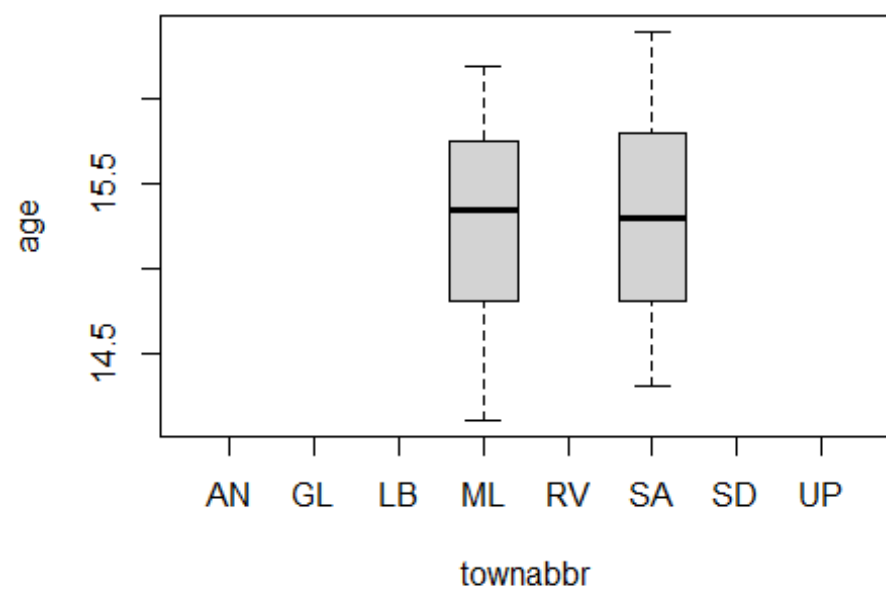
Continuous Vars = boxplots + t-test

PM2.5 by Towns

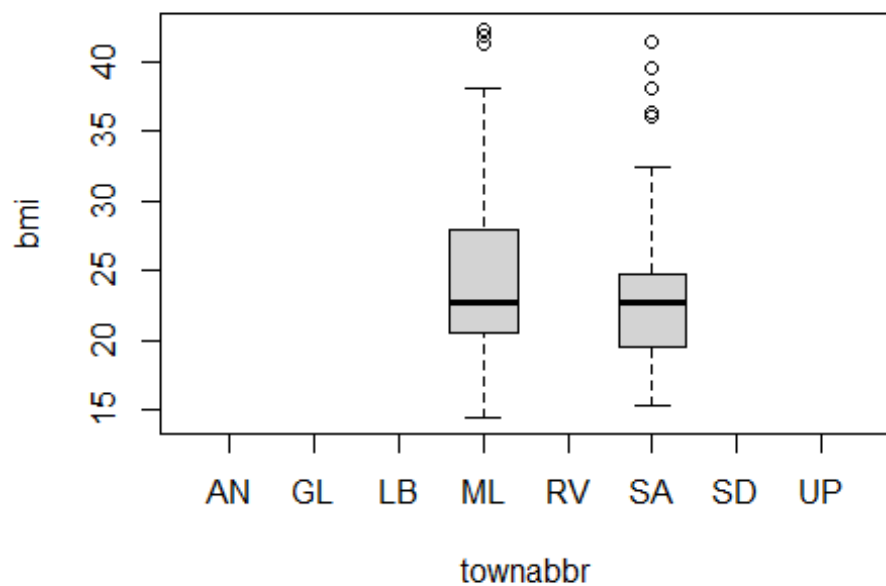




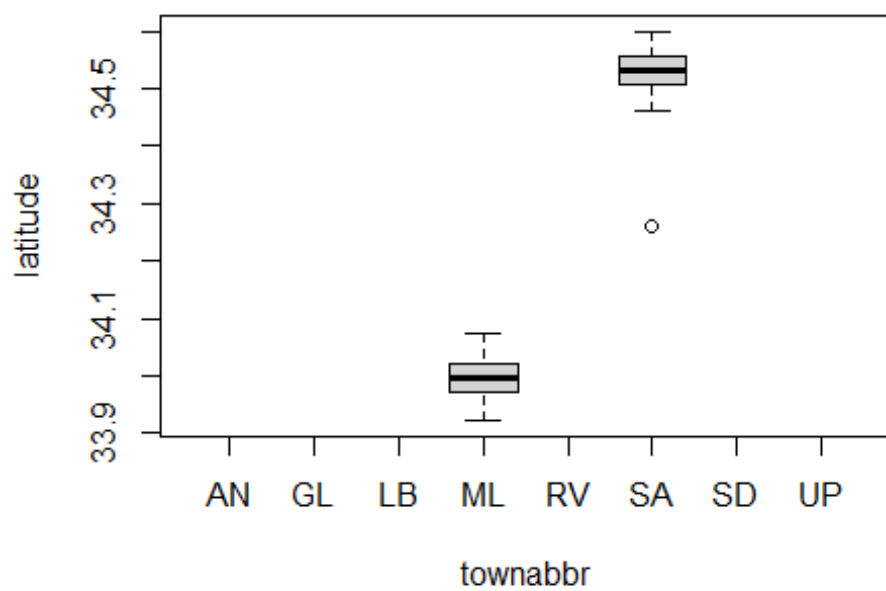
```
boxplot(age ~ townabbr, data=chs_data2)
```



```
boxplot(bmi ~ townabbr, data=chs_data2)
```



```
boxplot(latitude ~ townabbr, data=chs_data2)
```



```
t.test(pm25 ~ townabbr, data=chs_data2)
```

```

##
## Welch Two Sample t-test
##
## data:  pm25 by townabbr
## t = 137.38, df = 172.74, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group ML and group SA is not equal to 0
## 95 percent confidence interval:
##  8.615103 8.866265
## sample estimates:
## mean in group ML mean in group SA
##      17.43728      8.69660

t.test(nitrate ~ townabbr, data=chs_data2)

##
## Welch Two Sample t-test
##
## data:  nitrate by townabbr
## t = 54.18, df = 125.21, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group ML and group SA is not equal to 0
## 95 percent confidence interval:
##  0.8814341 0.9482691
## sample estimates:
## mean in group ML mean in group SA
##      2.775465      1.860613

t.test(age ~ townabbr, data=chs_data2)

##
## Welch Two Sample t-test
##
## data:  age by townabbr
## t = -0.66595, df = 198.88, p-value = 0.5062
## alternative hypothesis: true difference in means between group ML and group SA is not equal to 0
## 95 percent confidence interval:
## -0.2125994 0.1052561
## sample estimates:
## mean in group ML mean in group SA
##      15.24239      15.29606

t.test(bmi ~ townabbr, data=chs_data2)

##
## Welch Two Sample t-test
##
## data:  bmi by townabbr
## t = 1.9395, df = 171.9, p-value = 0.05407
## alternative hypothesis: true difference in means between group ML and group

```

```

p SA is not equal to 0
## 95 percent confidence interval:
## -0.02523194 2.87757702
## sample estimates:
## mean in group ML mean in group SA
##      24.49783      23.07165

t.test(latitude ~ townabbr, data=chs_data2)

##
## Welch Two Sample t-test
##
## data: latitude by townabbr
## t = -104.94, df = 211.35, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group ML and group SA is not equal to 0
## 95 percent confidence interval:
## -0.5435241 -0.5234811
## sample estimates:
## mean in group ML mean in group SA
##      33.99759      34.53109

## this should tell us whether SA and ML are significantly different for each variable

## Binary Variable (male) vs towns -> Proportion test

table(chs_data2$townabbr, chs_data2$male)

##
##      0  1
## AN  0  0
## GL  0  0
## LB  0  0
## ML 46 46
## RV  0  0
## SA 63 64
## SD  0  0
## UP  0  0

prop.table(table(chs_data2$townabbr, chs_data2$male), margin=1)

##
##      0      1
## AN
## GL
## LB
## ML 0.500000 0.500000
## RV
## SA 0.496063 0.503937

```

```

## SD
## UP

##since trying to filter the data to a chi-square test, the possible for an incorrect answer kept appearing. So we will now try to use the Fisher's Exact Test since its a smaller data set now that we are only looking at 2 towns and not all 8(i think its 8 lol)

fisher.test(table(chs_data2$townabbr, chs_data2$male))

##
## Fisher's Exact Test for Count Data
##
## data: table(chs_data2$townabbr, chs_data2$male)
## p-value = 1
## alternative hypothesis: two.sided

## Now we test Correlation WITHIN each town (the good stuff)

## First we split data by town
TownA <- subset(chs_data2, townabbr == "ML")
TownB <- subset(chs_data2, townabbr == "SA")

## Now we run a Correlation Matrix (with ONLY continuous vars)
varsC <- c("pm25", "bmi", "age", "nitrate", "latitude")
cor(TownA[, varsC], use="complete.obs")

##
##          pm25          bmi          age          nitrate          latitude
## pm25      1.00000000  0.22177181 -0.023149121  0.944760281 -0.35980908
## bmi       0.22177181  1.00000000  0.047544060  0.243963635 -0.08650636
## age      -0.02314912  0.04754406  1.000000000 -0.003780499  0.02121214
## nitrate   0.94476028  0.24396363 -0.003780499  1.000000000 -0.42553827
## latitude -0.35980908 -0.08650636  0.021212144 -0.425538270  1.00000000

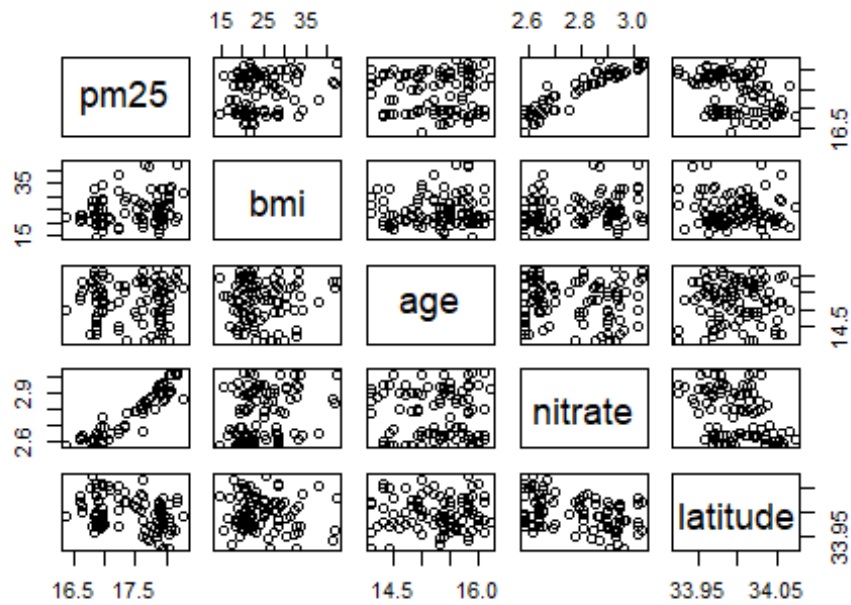
cor(TownB[, varsC], use="complete.obs")

##
##          pm25          bmi          age          nitrate          latitude
## pm25      1.00000000  0.02801673 -0.10363689 -0.03111258 -0.54195943
## bmi       0.02801673  1.00000000  0.03407113  0.11654554 -0.07966587
## age      -0.10363689  0.03407113  1.00000000  0.14669635 -0.04062683
## nitrate  -0.03111258  0.11654554  0.14669635  1.00000000 -0.04234236
## latitude -0.54195943 -0.07966587 -0.04062683 -0.04234236  1.00000000

## Add Visual Correlation (for better interpretation)
pairs(TownA[, varsC], main="ML Relationships")

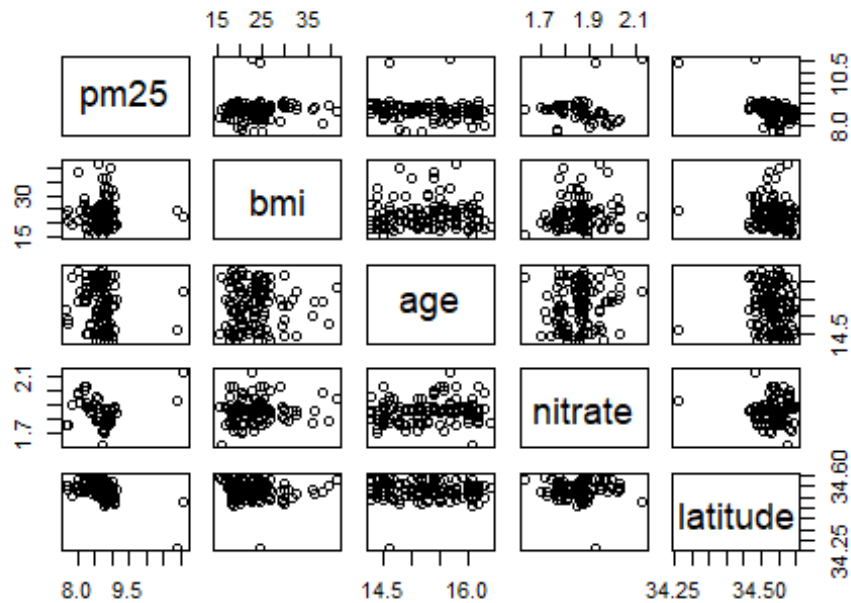
```

ML Relationships



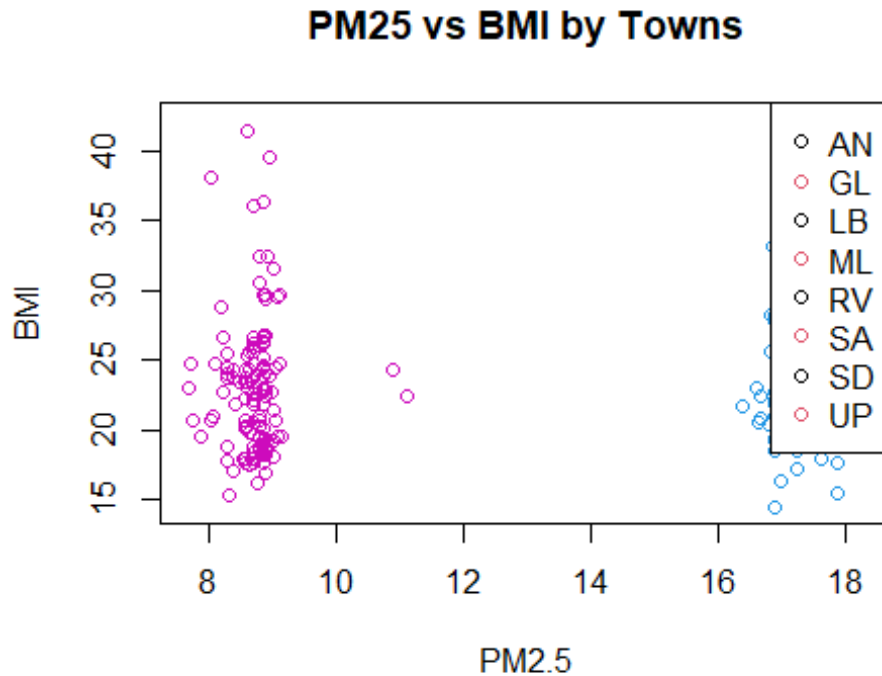
```
pairs(TownB[, varsC], main="SA Relationships")
```

SA Relationships



I now would like to see if Scatter plots by Town vs each variable would be better for interpretation. We start with PM2.5 vs BMI by Town then, Nitrate vs BMI by Town, PM2.5 vs Age, and finally Nitrate vs Age by Town

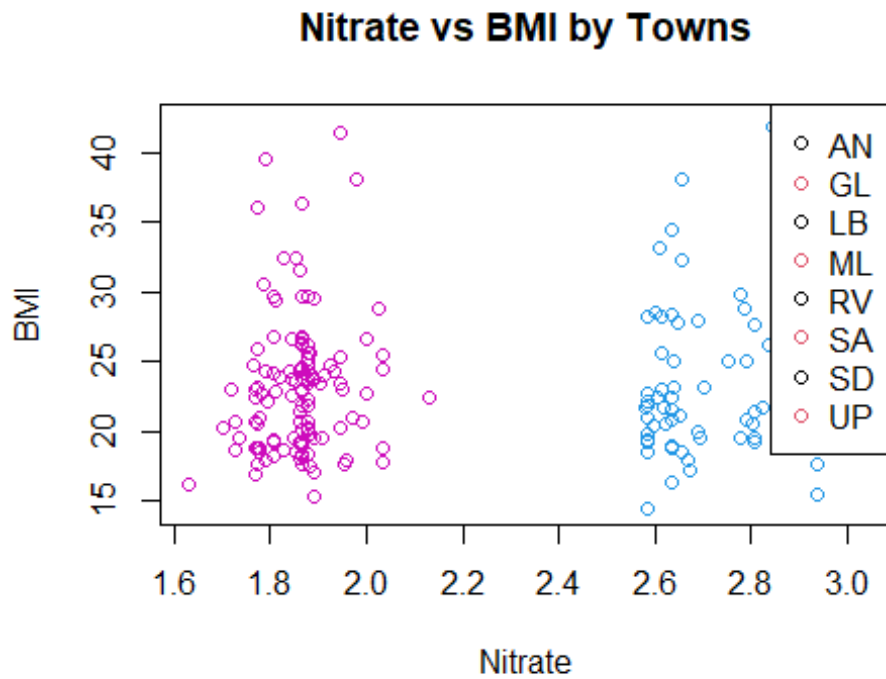
```
plot(chs_data2$pm25, chs_data2$bmi, col = as.factor(chs_data2$townabbr),
     main = "PM25 vs BMI by Towns",
     xlab = "PM2.5", ylab = "BMI")
legend("topright", legend=levels(chs_data2$townabbr), col=1:2,
      pch=1)
```



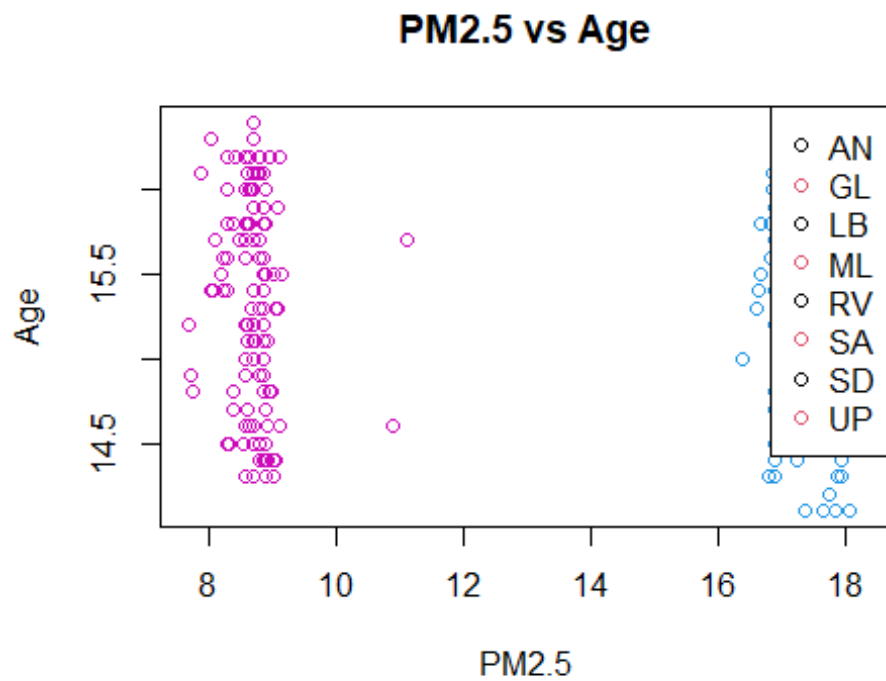
in plot(...) above I wrap the Column in as.factor() to tell R to treat the abbreviations in the code as categories and assign a color index to each

Now we continue

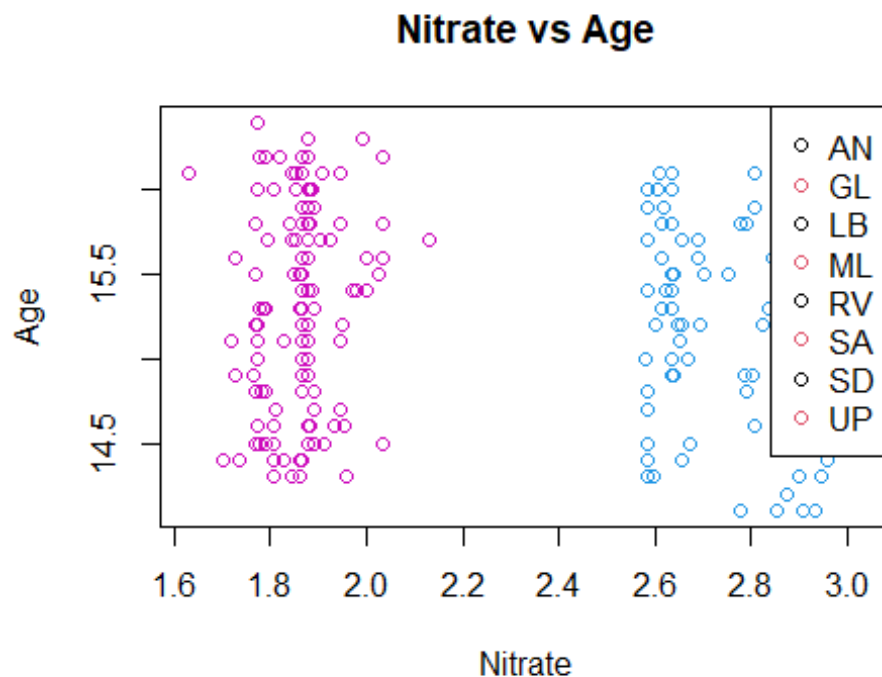
```
plot(chs_data2$nitrate, chs_data2$bmi, col =
as.factor(chs_data2$townabbr),
main = "Nitrate vs BMI by Towns",
xlab = "Nitrate", ylab = "BMI")
legend("topright", legend=levels(chs_data2$townabbr), col=1:2,
      pch=1)
```



```
plot(chs_data2$pm25, chs_data2$age, col =
      as.factor(chs_data2$townabbr),
      main = "PM2.5 vs Age",
      xlab = "PM2.5", ylab = "Age")
legend("topright", legend=levels(chs_data2$townabbr), col=1:2,
      pch=1)
```



```
plot(chs_data2$nitrate, chs_data2$age, col =
      as.factor(chs_data2$townabbr),
      main = "Nitrate vs Age",
      xlab = "Nitrate", ylab = "Age")
legend("topright", legend=levels(chs_data2$townabbr), col=1:2,
      pch=1)
```



The last analysis I'll try is actually combining into regression to ask: Does pollution affect outcome differently by town?

```
model <- lm(bmi ~ pm25 * townabbr + nitrate * townabbr + age + male + latitude, data = chs_data2)
summary(model)
```

```
##
## Call:
## lm(formula = bmi ~ pm25 * townabbr + nitrate * townabbr + age +
##     male + latitude, data = chs_data2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.259  -3.574  -0.894   2.159  17.482
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    125.4878    367.1112   0.342   0.733
## pm25           -0.7389     3.3202  -0.223   0.824
## townabbrSA      5.6279    35.5622   0.158   0.874
## nitrate         11.9404    11.2869   1.058   0.291
## age             0.2551     0.5998   0.425   0.671
## male           -1.0137     0.7069  -1.434   0.153
## latitude       -3.6658    10.7282  -0.342   0.733
## pm25:townabbrSA  1.1235     3.5955   0.312   0.755
## townabbrSA:nitrate -5.5969    12.7881  -0.438   0.662
##
```

```
## Residual standard error: 5.159 on 210 degrees of freedom
## Multiple R-squared:  0.06647,    Adjusted R-squared:  0.03091
## F-statistic: 1.869 on 8 and 210 DF,  p-value: 0.06633
```

The code above tells us that pm25::TownB (estimate: 1.1235, p =0.755) -> the effect of PM2.5 on BMI is slightly more positive in town SA than the reference town, making the difference insignificant. TownB::Nitrate (estimate: -5.5969, p = 0.662) -> the effect of nitrate on BMI is weaker (more negative) in town SA, making the difference insignificant.

The main pollutant effects: the values im about to explain represent the effect of the pollutants specifically in the reference town (ML not SA). PM2.5 (estimate: -0.7389, p = 0.824) -> Higher PM2.5 is tied to a tiny decrease in BMI, this is insignificant. Nitrate (estimate: 11.9404, p = 0.291) -> Higher nitrate is tied to an increase in BMI, this is insignificant.

Overall R-squared(0.0309) -> this model only explains about 3% of the variance in BMI. Model p-value(0.0663) -> the overall model is not significant at the standard 5% level, though it shows marginal significance at the 10% level. So we shall run more test to find significance within our towns and variables.

after running these test we can now clearly see that there are no statistically significant association between PM2.5 or Nitrate exposure and BMI after adjusting for demographic variables. In addition, interaction terms between town and pollutant levels were not significant which suggests that the relationship between pollution & BMI does not differ across towns. This is very normal considering BMI is not strongly driven by air pollution ; its more driven by diet, lifestyle, & genetics.

While our models aren't wrong, it's just not the best outcome variable. So we now compare a way stronger variable FEV1 for more real effects, meaningful correlations, and better R squared

```
lm(fev1~ pm25 * townabbr + nitrate * townabbr + age + male + latitude, data = chs_data2)
```

```
##
## Call:
## lm(formula = fev1 ~ pm25 * townabbr + nitrate * townabbr + age +
##      male + latitude, data = chs_data2)
##
## Coefficients:
##      (Intercept)                pm25                townabbrSA                nit
rate
##      12402.1                -577.3                -4353.0                24
27.6
##              age              male              latitude      pm25:townab
brSA
##      337.0                899.6                -323.1                6
90.3
```

```
## townabbrSA:nitrate
## -2273.2

## INFO still up for interpretation

## Now we are switching our Models to FEV1 instead of BMI

model_fev1 <- lm(fev1 ~ pm25 * townabbr + nitrate * townabbr + age + male + latitude,
                 data = chs_data2)
summary(model_fev1)

##
## Call:
## lm(formula = fev1 ~ pm25 * townabbr + nitrate * townabbr + age + male + latitude, data = chs_data2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1603.24  -352.49   -42.58   363.56  1369.70
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   12402.06   40658.13    0.305   0.7606
## pm25          -577.31    367.72   -1.570   0.1179
## townabbrSA   -4352.96    3938.57   -1.105   0.2703
## nitrate       2427.65    1250.04    1.942   0.0535 .
## age           337.03     66.43    5.074 8.56e-07 ***
## male          899.61     78.29   11.491 < 2e-16 ***
## latitude     -323.09    1188.17   -0.272   0.7859
## pm25:townabbrSA  690.34     398.21    1.734   0.0845 .
## townabbrSA:nitrate -2273.19   1416.30   -1.605   0.1100
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 571.4 on 210 degrees of freedom
## Multiple R-squared:  0.4466, Adjusted R-squared:  0.4256
## F-statistic: 21.19 on 8 and 210 DF,  p-value: < 2.2e-16

## The code first test whether the impact of pollutants on Lung function (fev1) differs between town SA & ML. pm25:townabbrSA (estimate: 690.34, p-value = 0.0845) -> this is marginally significant at the 10% level (indicated by the . code), which suggests that the negative effect of PM2.5 on Lung function is weakened (moved in the positive direction by 690.34 unit) specifically in SA compared to ML. townabbrSA:nitrate (estimate: -2273.19, p = 0.1100) -> this interaction is not statistically significant, there is no strong evidence that the effect of nitrate differs between SA & ML.

## The main pollutant effects values represent the impact of the pollutants specifically within SA and ML. nitrate (estimate: 2427.65, p = 0.0535) -> this
```

is Marginally significant at the 10% level (.). In ML, higher nitrate levels are associated with a baseline increase in FEV1. pm25 (estimate: -577.31, $p = 0.1179$) -> while direction is negative (higher PM2.5 relates to lower lung function), this effect is not statistically significant at standard thresholds

Town Differences: townabbrSA (estimate: -4352.96, $p = 0.2703$) -> there is no statistically significant difference in baseline FEV1 levels between town SA and ML when all pollutant levels are at 0.

the strongest drivers of lung function in this dataset are demographic factors, not the environment: male (estimate: 899.61, $p < 2 \times 10^{-16}$) -> Highly significant, males have significantly higher FEV1 levels than females by roughly 900 units. age (estimate: 337.03, $p = 8.56 \times 10^{-7}$) -> Highly Significant, for every year of age FEV1 increases by 337 units.